

Matrix functions and stability

The uniform expansion threshold for Toeplitz symbols with higher-order zeros

Difficulty: challenging **Importance:** interesting to specialist**Status:** [Lean verified](#)

Rating rationale: Matching bulk asymptotics with extreme eigenvalues at the exact breakdown order is challenging; the detailed expansion threshold is mainly important to structured spectral specialists.

Lean proof and verification evidence — 20 September 2026

The [complete formalization](#) proves all three original assertions for every $m \geq 3$, the obstruction for every continuous coefficient family, and the revised proof's smooth common family with matching upper and finite-head lower error bounds. The exact Fourier-integral matrix correspondence and coefficient uniqueness are separately exported. No analytic or transcendence hypothesis remains unproved in these results.

The immutable local proof commit `e378ec4679f9119aac79a0fe30b80493a61ec5b` passed the actual non-root Linux Comparator, Lean default-kernel replay, standard-axiom checks, and real sandbox/rejection controls. [Checked input snapshot and raw evidence](#) · [Five proved exports](#) · [Exact statement correspondence](#).

Separate nonimplementing AI agents approved the [formal statement and source correspondence](#) and [proof and runtime evidence](#); an additional [scoped cross-review](#) records authorship exclusions. The formalization and September proof revisions were produced by OpenAI Codex AI agents. George Stepaniants retains original manuscript authorship, and the conjecture and simple-loop method retain the cited authors' attribution. No external human peer review or source-author endorsement is claimed. Verification was completed locally before the authorized publication in [PR #309](#).

Proof clarification - 19 September 2026

The [revised proof](#) explicitly proves uniqueness of the continuous coefficient functions on the logarithmic-squared bulk grid (Lemma 5, following Proposition 4.2 of Barrera–Böttcher–Grudsky–Maximenko). This identifies any hypothetical all-index expansion with the constructed one. Equation (25) supplies only an upper bound; the obstruction uses the separate inverse-trace contradiction in equations (26)–(33). Corollary 6 now gives a matching lower bound of order h^{2m} for the maximum error over a fixed finite set of low indices.

The revision also credits the simple-loop method explicitly and adds Bogoya–Grudsky (2025). The [dated audit](#) records the mathematical checks, source comparison, and coefficient-uniqueness repair. The [initial Lean audit](#) records the earlier partial development; it is superseded by the completed verification above. The original problem, permanent ID, grid, cutoff, and quantifiers below are unchanged.

Affirmative resolution - 12 September 2026 (UTC)

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[Theorem 1](#) and [Sections 2-5 of the complete proof](#) establish all three assertions below for every integer $m \geq 3$, using the same smooth coefficient functions with $d_0 = g_m$. The expansion is uniform over every eigenvalue through order $2m - 1$, extends to order $2m$ above the stated logarithmic-squared index cutoff, and fails to extend uniformly to all indices at that order. The matrix, grid normalization, parameter quantifiers and cutoff are unchanged.

[Proof PDF](#) · [Standalone proof TeX](#) · [Independent mathematical review](#) · [Submission and public-source audit](#).

The original proof passed a separate Codex-agent informal audit. It derived the bulk expansion from an exact boundary determinant and the final obstruction from a classical uniform inverse-kernel limit and a trace identity; the revision above supplies a direct finite-matrix trace calculation. Barrera, Böttcher, Grudsky, Maximenko and the cited later authors retain credit for the conjecture and prior cases; Böttcher-Widom and their cited predecessors retain credit for the inverse-kernel theorem. Substantial AI assistance is disclosed. That initial audit was informal; the completed Lean verification is documented above. External human peer review is not claimed. The original statement, permanent ID, references and dated history are retained below; the ratings are historical.

Statement

For each integer $m \geq 3$, let

$$g_m(\theta) = (2 \sin(\theta/2))^{2m}, \quad \hat{g}_{m,k} = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_m(\theta) e^{-ik\theta} d\theta.$$

Let $T_n(g_m) = (\hat{g}_{m,j-k})_{j,k=1}^n$, with eigenvalues $\lambda_{n,1} \leq \dots \leq \lambda_{n,n}$ counted with multiplicity. For real continuous functions $d_0, \dots, d_{2m}$ on $[0, \pi]$, independent of n, j , write

$$R_{p,n,j} = \lambda_{n,j} - \sum_{k=0}^p \frac{d_k(j\pi/(n+2))}{(n+2)^k}.$$

Conjecture 8.4 of Barrera–Böttcher–Grudsky–Maximenko. Such coefficient functions exist, with $d_0 = g_m$, for which all three assertions hold:

1. For every integer $0 \leq p \leq 2m - 1$, there are constants $D_p > 0$ and $N_p \in \mathbb{N}$ such that

$$|R_{p,n,j}| \leq D_p (n+2)^{-p-1} \quad (n \geq N_p, 1 \leq j \leq n).$$

2. There are constants $D_{2m} > 0$ and $N_{2m} \in \mathbb{N}$ such that

$$|R_{2m,n,j}| \leq D_{2m} (n+2)^{-2m-1}$$

whenever $n \geq N_{2m}$ and $\lceil (\log(n+2))^2 \rceil \leq j \leq n$.

3. No constants $D > 0, N \in \mathbb{N}$ make the bound $|R_{2m,n,j}| \leq D(n+2)^{-2m-1}$ valid for every $n \geq N$ and every $1 \leq j \leq n$.

All constants may depend on m and the expansion order; $\log$ denotes the natural logarithm. The same coefficient functions are used throughout. The continuity and existence quantifiers make explicit the regular-expansion convention of the source's Theorem 1.2 and §8; these are not matrix-size-dependent fitted coefficients.

Numerical significance

Regular eigenvalue expansions support accurate computation without forming large Toeplitz matrices. This conjecture identifies exactly where extreme eigenvalues obstruct a uniform expansion for these polynomial symbols. It retains all three parts as one problem.

References and status check

- M. Barrera, A. Böttcher, S. M. Grudsky and E. A. Maximenko, *Eigenvalues of even very nice Toeplitz matrices can be unexpectedly erratic*, Oper. Theory Adv. Appl. **268** (2018), 51–77, [DOI](#); [author preprint](#), Conjecture 8.4, p. 26, equation (8.4); Theorem 1.2 supplies the proved $m = 2$ analogue.
- M. Barrera, S. Grudsky, V. Stukopin and I. Voronin, *Asymptotics of the eigenvalues of seven-diagonal Toeplitz matrices of a special form*, Adv. Oper. Theory **9** (2024), 79, [DOI](#); [preprint](#), Theorems 2.3–2.6. This studies the sixth-order-zero case with more elaborate formulas.

- M. Bogoya and S. Grudsky, *Eigenvalues of non-Hermitian banded Toeplitz matrices approaching simple points of the limiting set*, *Comput. Math. Math. Phys.* **65** (2025), 1453–1471, DOI; [author-hosted paper](#), Theorems 2.1–2.2. This gives local phase equations and expansions near nondegenerate simple points; the revised proof records the methodological connection without a novelty claim.
- A. Böttcher, *Ten years with Sergei Grudsky in the eigenvalue bulk of Toeplitz matrices*, *J. Math. Sci.* **298** (2026), 363–376, DOI, section “Beyond the simple-loop class,” Theorems 2–5 and discussion of higher-order zeros.

On 2026-09-10, checked the original full preprint, the seven-diagonal follow-up’s full preprint and publication record, the 2026 survey, and targeted title/conjecture/2025–2026 searches. No resolution of all parts for every $m \geq 3$ was located. Later local second-order expansions and results confined to $m = 3$ do not establish this full statement. This bounded search is not a proof of openness.

Audit — 2026-09-10

Independently rechecked [Conjecture 8.4](#), [the seven-diagonal paper’s Theorems 2.3–2.6](#), and [Böttcher’s survey](#). Corrected the status to Open: the proved $m = 2$ analogue lies outside the displayed range, and the $m = 3$ local and second-order formulas do not establish the three-part threshold assertion. The [March 2026 eigenvalue-superposition paper](#) assumes simple-loop symbols, excluding these higher-order zeros. Title and conjecture searches found no complete resolution. Challenging difficulty and specialist importance are retained.

Resolution audit - 2026-09-12

The independent audit checked the complete all- m argument, including the coalescing characteristic roots, normalized determinant remainder, upper-endpoint cancellation, eigenvalue indexing, all three expansion orders, and dominated trace limit. The primary-source and public branch/fork/PR checks found no existing full resolution at the recorded time; the linked submission record gives the precise scope and limitations of that search.